

$$P(E^c) = 1 - P(E)$$

$$P(A) = P(A \cap B) + P(A - B)$$

$$\text{De Morgan's: } \overline{(E \cup F)} = \bar{E} \cap \bar{F} \quad \binom{n}{k} = \frac{n!}{k! (n-k)!}$$

$$\overline{(E \cap F)} = \bar{E} \cup \bar{F}$$

$$\text{Classical: } \frac{\text{desired}}{\text{total}}$$

$$0 \leq P(E) \leq 1, P(\emptyset) = 0$$

$$\text{disjoint: } P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

Conditional: new info shrinks sample space

$P(E|F)$: probability of E given F

$$0 \leq P(E|F) \leq 1, P(\emptyset|F) = 0$$

$$\text{disjoint: } P(E_1 \cup E_2 | F) = \sum_n P(E_n | F)$$

$$P(E|F) = \frac{P(EF)}{P(F)}$$

$$P(EF) = P(E|F)P(F) = P(F|E)P(E)$$

Partition Rule

$$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c)$$

$$P(B) = \sum P(B|A_n)P(A_n)$$

→ conditional

$$P(B|E) = P(B|A_1 E)P(A_1|E) + \dots + P(B|A_n E)P(A_n|E)$$

Chain Rule: things happening in a row

$$P(A_1, A_2, A_3) = P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1, A_2)$$

$$\rightarrow P(A_1, A_2, \dots, A_n) = P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1, A_2) \cdot \dots \cdot P(A_n|A_1, A_2, \dots, A_{n-1})$$

Bayes Formula:

$$P(A_i|B) = \frac{P(B|A_i) P(A_i)}{\sum_{j=1}^n P(B|A_j) P(A_j)}$$

slipped: $P(B|A_i) \leftrightarrow P(A_i|B)$

$$(\text{prob of cause } A) = \frac{\text{prob of result given cause } A}{\text{total prob of result from all causes}}$$

Events E, F are independent if $P(EF) = P(E)P(F)$

$$P(F) \neq 0 \rightarrow P(E|F) = P(E)$$

$\rightarrow$ statistical independence: information, not physical interaction

E is ind. of $F \rightarrow$ learning that F has occurred gives no new info to help predict if E will occur

if E, F are ind., then 1) E, F' are ind.

$$\leftrightarrow E', F \text{ are ind.}$$

$$E', F' \text{ are ind.}$$

(if knowing F happened doesn't help predict E , then knowing F didn't happen doesn't help predict E either)

$$P(EF^c) = P(E)P(F^c|E)$$

disjoint events are not independent

independence of variables implies independence of events

Events A, B, C are independent if

1) $A \perp B$

2) $B \perp C$

3) $A \perp C$

4) $P(ABC) = P(A)P(B)P(C)$

$P(ABC) = P(A)P(B)P(C)$ does not imply pairwise independence

Conditional Independence

$A \perp B | C$ (conditionally independent given C) if

$$P(A|BC) = P(A|C)P(B|C) \text{ assuming } P(C) \neq 0$$

— if we know C has happened, learning B gives no info on A

$$A \perp B | C \iff P(A|BC) = P(A|C)$$

$$\hookrightarrow P(A-B) = P(A)(1-P(B))$$

$P(AB) < P(A)P(B) \rightarrow$ repulsion ($P(AB)=0 \rightarrow$ disjoint)

$P(AB) = P(A)P(B) \rightarrow$ independence

$P(AB) > P(A)P(B) \rightarrow$ attraction

geometric series: $\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}, |r| < 1$ Sum = $\frac{\text{1st term}}{1 - \text{ratio}}$

$$\sum_{k=0}^{n-1} ar^k = a \frac{1-r^n}{1-r}, r \neq 1$$

$$\sum_{k=1}^n ar^{k-1} = a \frac{1-r^n}{1-r}$$

Independent trials

P of getting K occurrences of E in n trials:

$$P(\text{exactly } k \text{ successes}) = \binom{n}{k} P(E)^k (1 - P(E))^{n-k}$$

$\binom{n}{k} = \binom{n}{n-k}$ if rolling 5 dice,
choosing 2 as success is the
same as choosing 3 as failures

$$\sum_{k=0}^n \binom{n}{k} p^k (1-p)^{n-k} = 1 \quad (\text{binomial theorem})$$

- if you add up the probability of getting

0, 1, 2, ..., n successes, it must total to 1

binomial distribution function

- probability of getting k successes in n independent trials, where each trial has the same probability of success p

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Random Variables

function that maps outcomes in a sample space to a real number

$$X: S \rightarrow \mathbb{R}$$

Let $A \subseteq \mathbb{R}$ ← event

$$\{X \in A\} = \{u \in S : X(u) \in A\}$$

u : a specific outcome

$X(u)$: scalar assigned to that outcome

CDF

CDF: cumulative distribution function

$$F_X(u) = P(X \leq u) \quad \text{for } -\infty < u < \infty$$

↑ which random var.

- tells accumulated probability $P(X \leq u)$

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

General : $F_X(u) = \sum_{k \leq u} p(x=k)$

Geometric Series : $F_X(u) = \sum_{k=1}^{\lfloor u \rfloor} (1-p)^{k-1} p$

$$\rightarrow F_N(u) = 1 - (1-p)^{\lfloor u \rfloor}$$